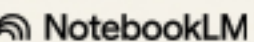
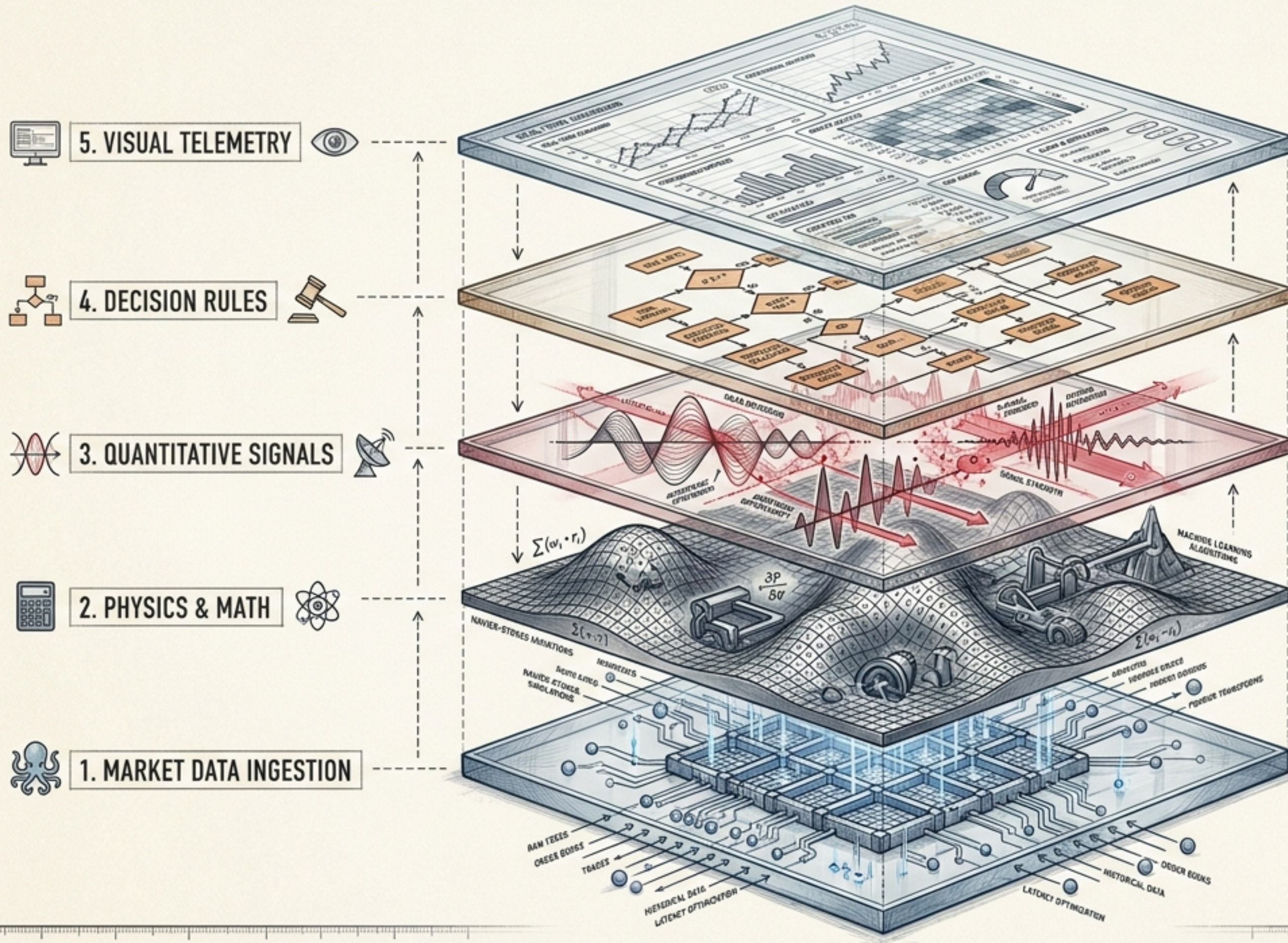


Deconstructing Symm:

A physics-inspired algorithmic trading architecture.

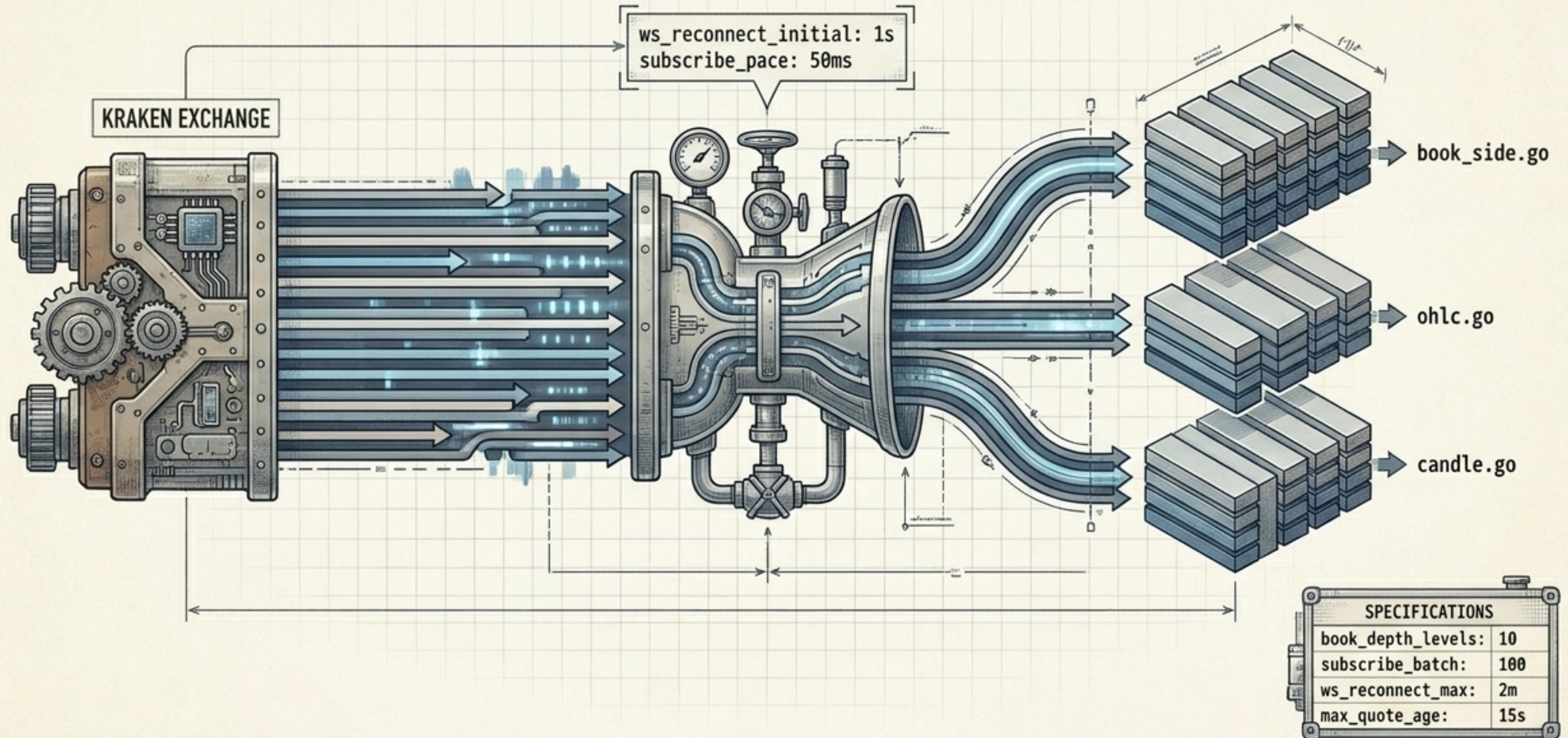


THE CHRONOLOGICAL PIPELINE OF A QUANT SYSTEM



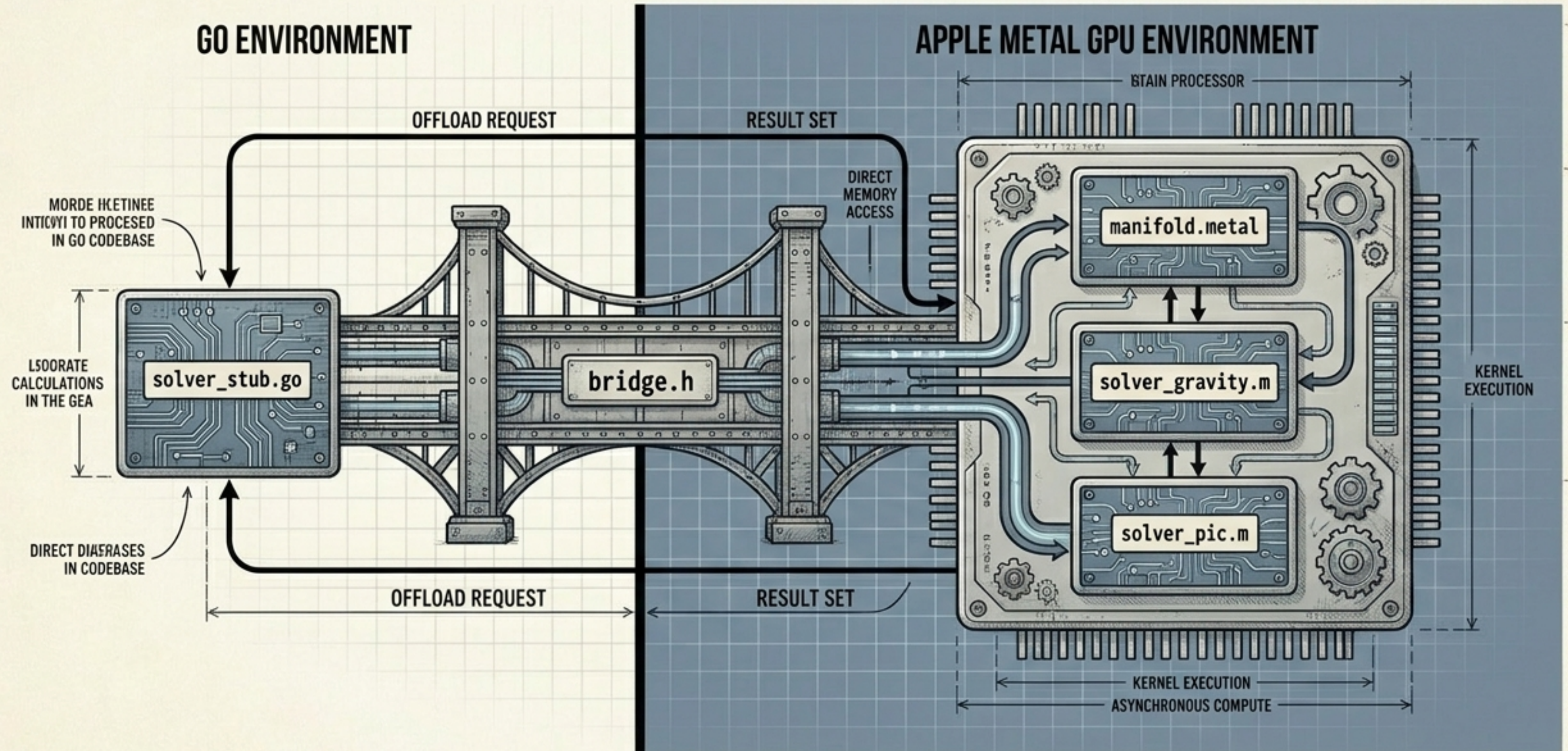
HIGH-FIDELITY MARKET DATA INGESTION

The **kraken** package establishes resilient L3 orderbook telemetry, dynamically managing websocket reconnections and batching updates to prevent dropped frames.



HARDWARE-ACCELERATED PHYSICS SOLVERS

Standard execution cannot handle continuous fluid models. The numeric/physics package offloads heavy matrix calculations directly to Apple Metal GPUs.

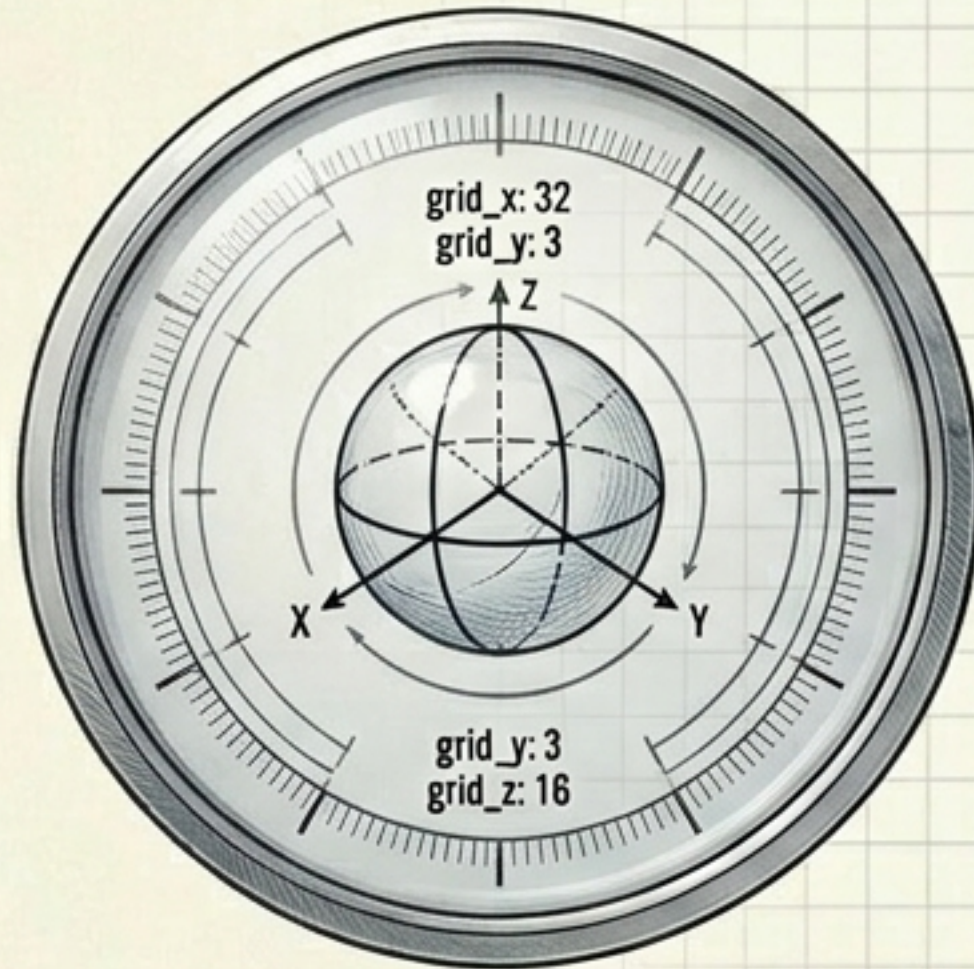


THE SIGNAL INTELLIGENCE MATRIX

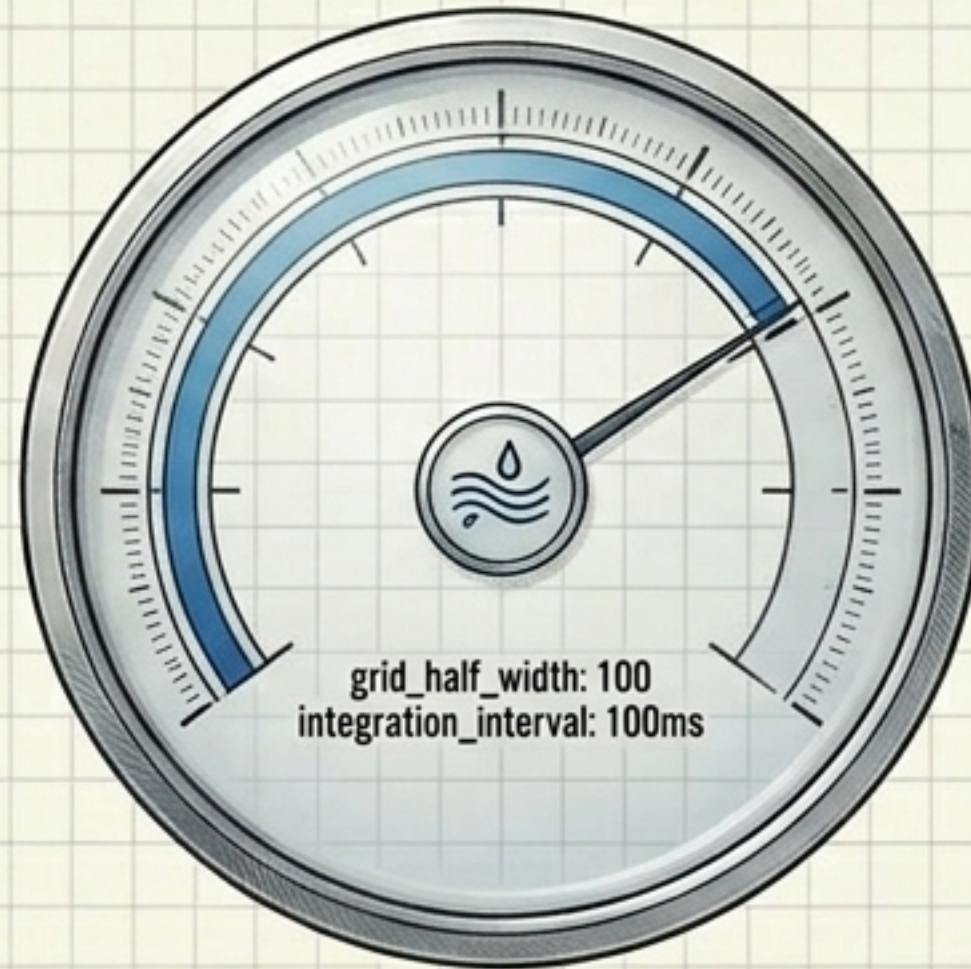
Signal	Basis	Input	Target Anomaly
Hawkes	Event Arrival Tracking	excitation_state.go	Micro-burst order clustering
Fluid/Manifold	Orderbook Physics	grid_velocity.go	Latent liquidity voids
Causal	Contagion Dynamics	cross_section.go	Cross-asset volatility bleeding
Toxicity	Adverse Selection	tracker.go	Informed taker flow

MODELING MARKET MICROSTRUCTURE AS CONTINUOUS SPACE

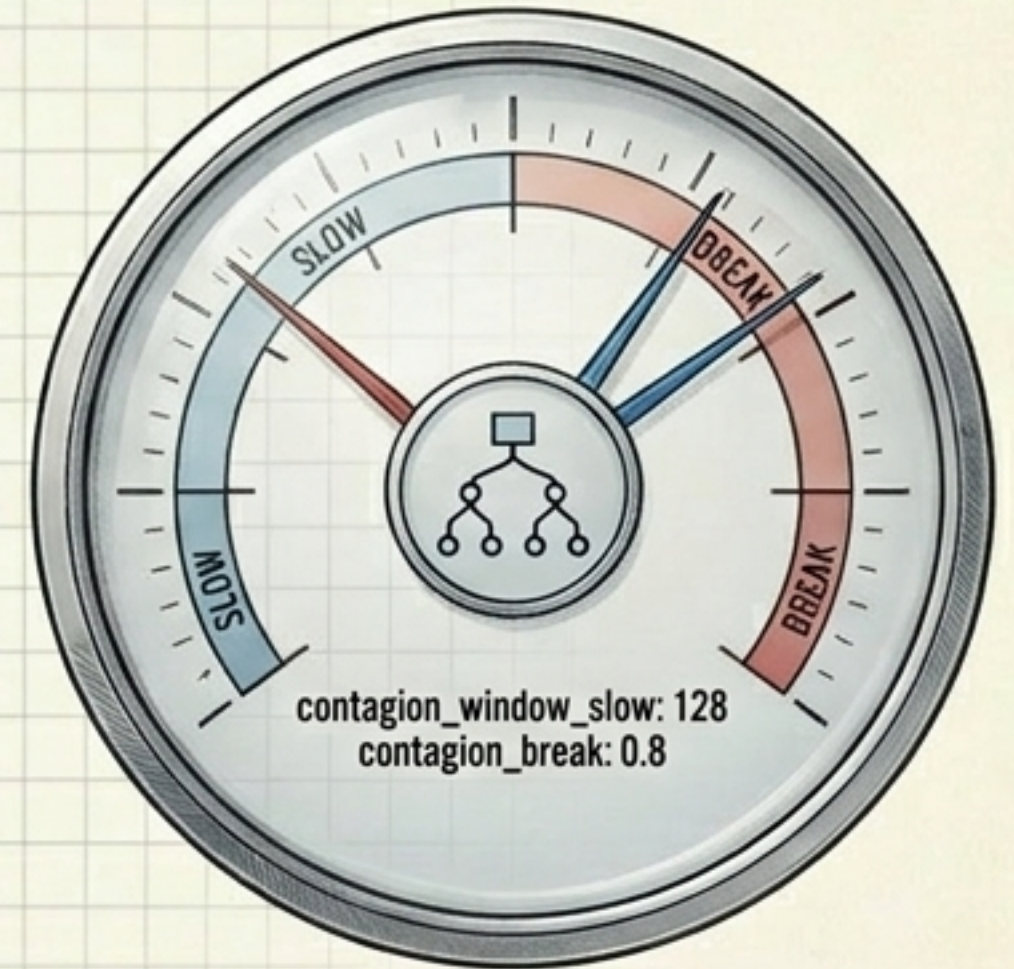
Symm abandons discrete price points, mapping the orderbook as a continuous 3D mathematical space where liquidity behaves like a fluid.



MANIFOLD



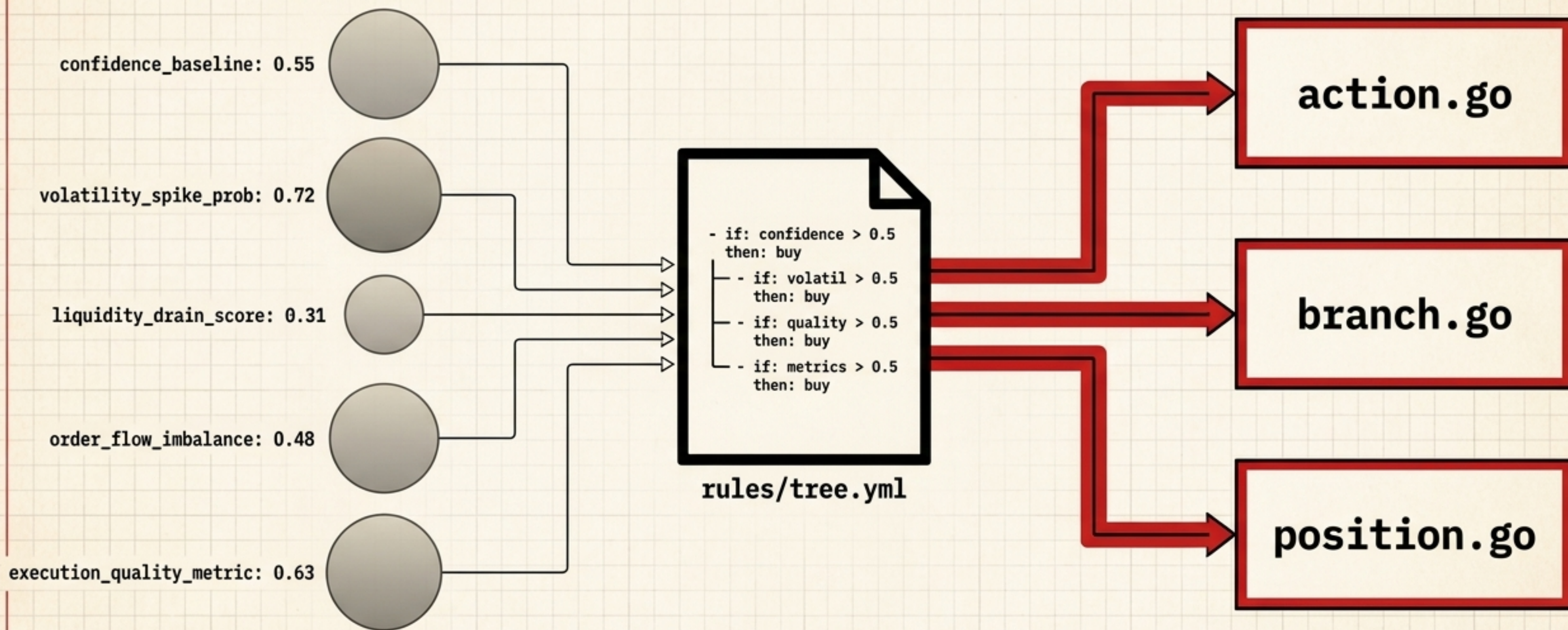
FLUID



CAUSAL

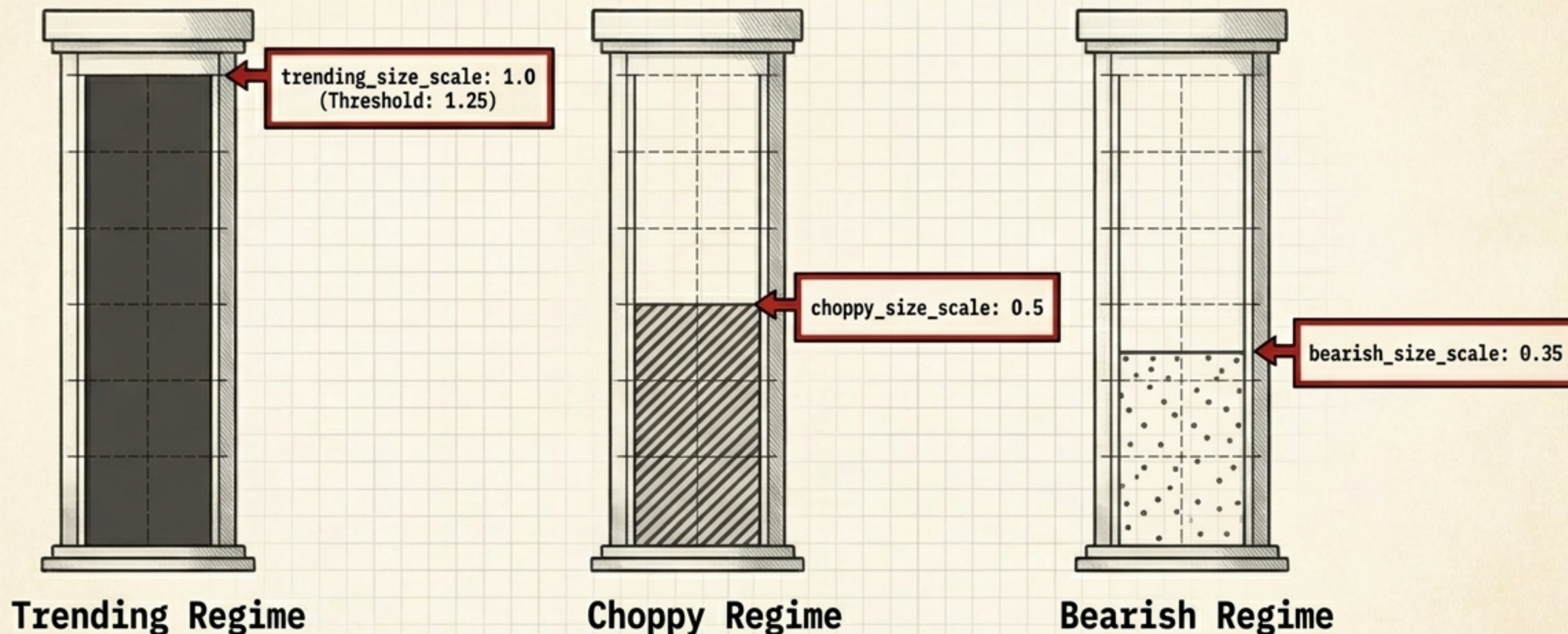
COLLAPSING PROBABILITIES INTO BINARY ACTIONS

Quantitative signals form a **continuous spectrum**. The **logic** package evaluates these states against a strict YAML tree to execute deterministic trades.



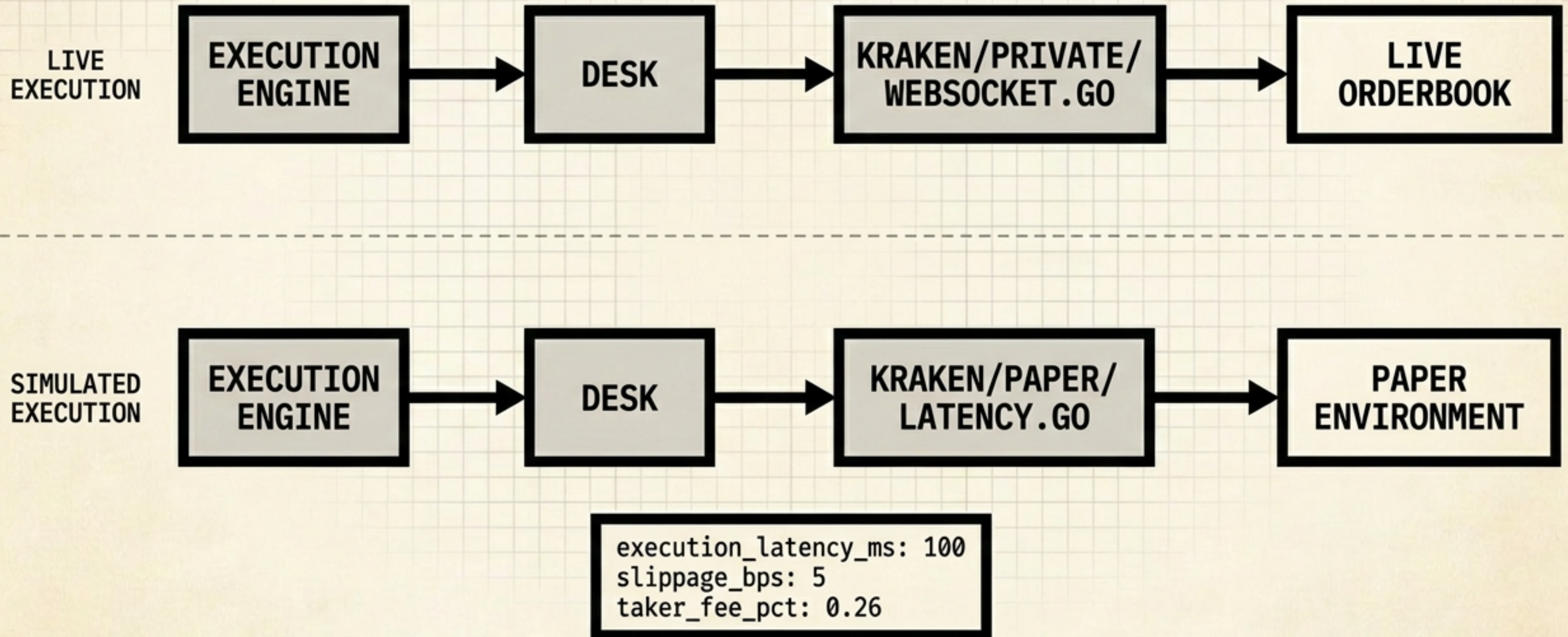
DYNAMIC RISK SCALING VIA REGIME ADAPTATION

Capital allocation **adapts dynamically**. The system tracks volatility and trend strength to automatically throttle position sizes.



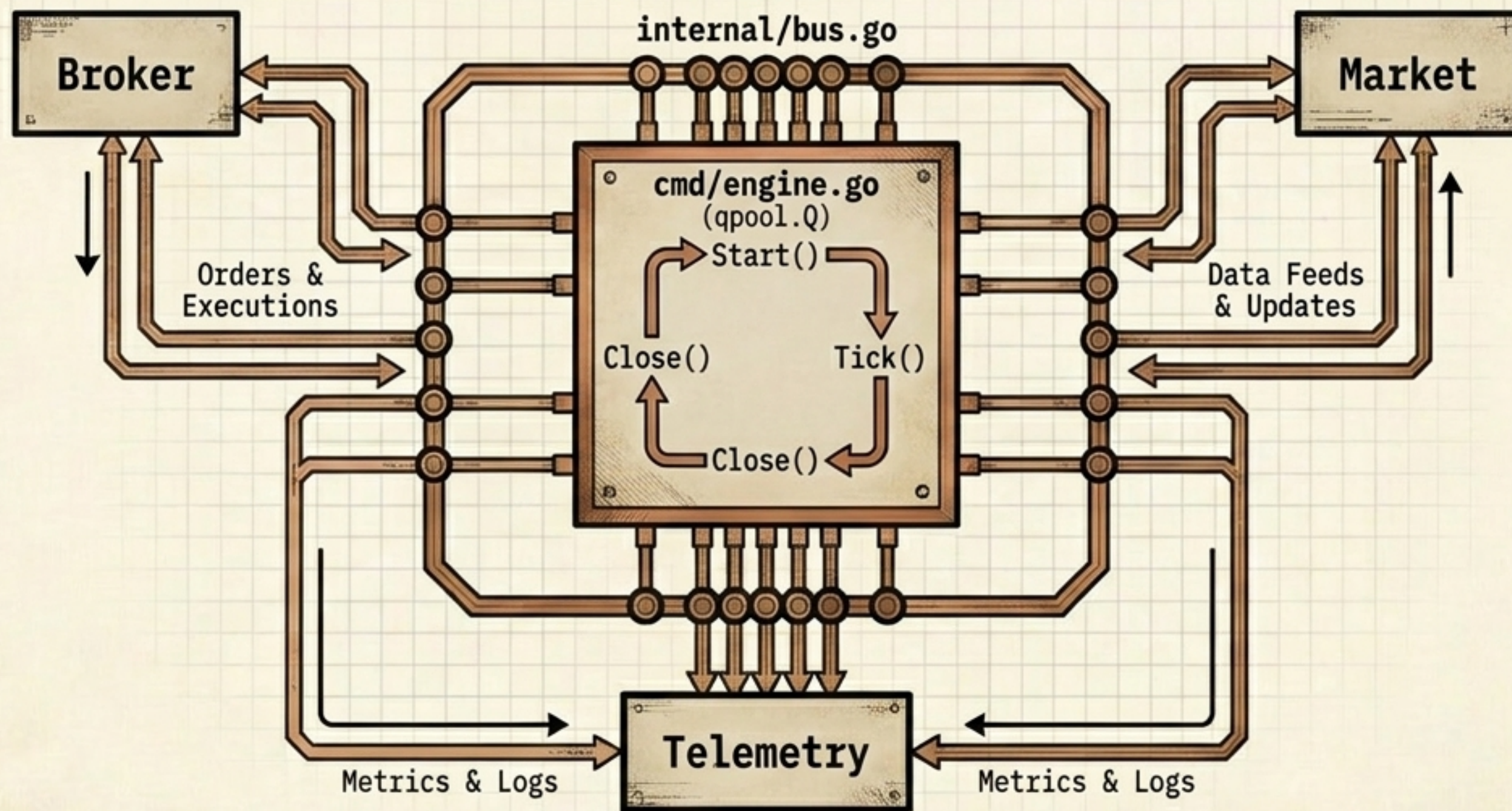
THE EXECUTION PIPELINE

The **broker/desk.go** engine isolates logic from mechanics, allowing flawless switching between live websockets and latency-injected simulated environments.



CONCURRENCY AND EVENT ORCHESTRATION

A **decoupled, context-aware Go Engine**. Systems communicate strictly via an internal message bus, ensuring high throughput without race conditions.



High-Throughput, Race-Free Communication via Message Bus.

OPTIMIZATION AND WALK-FORWARD VALIDATION

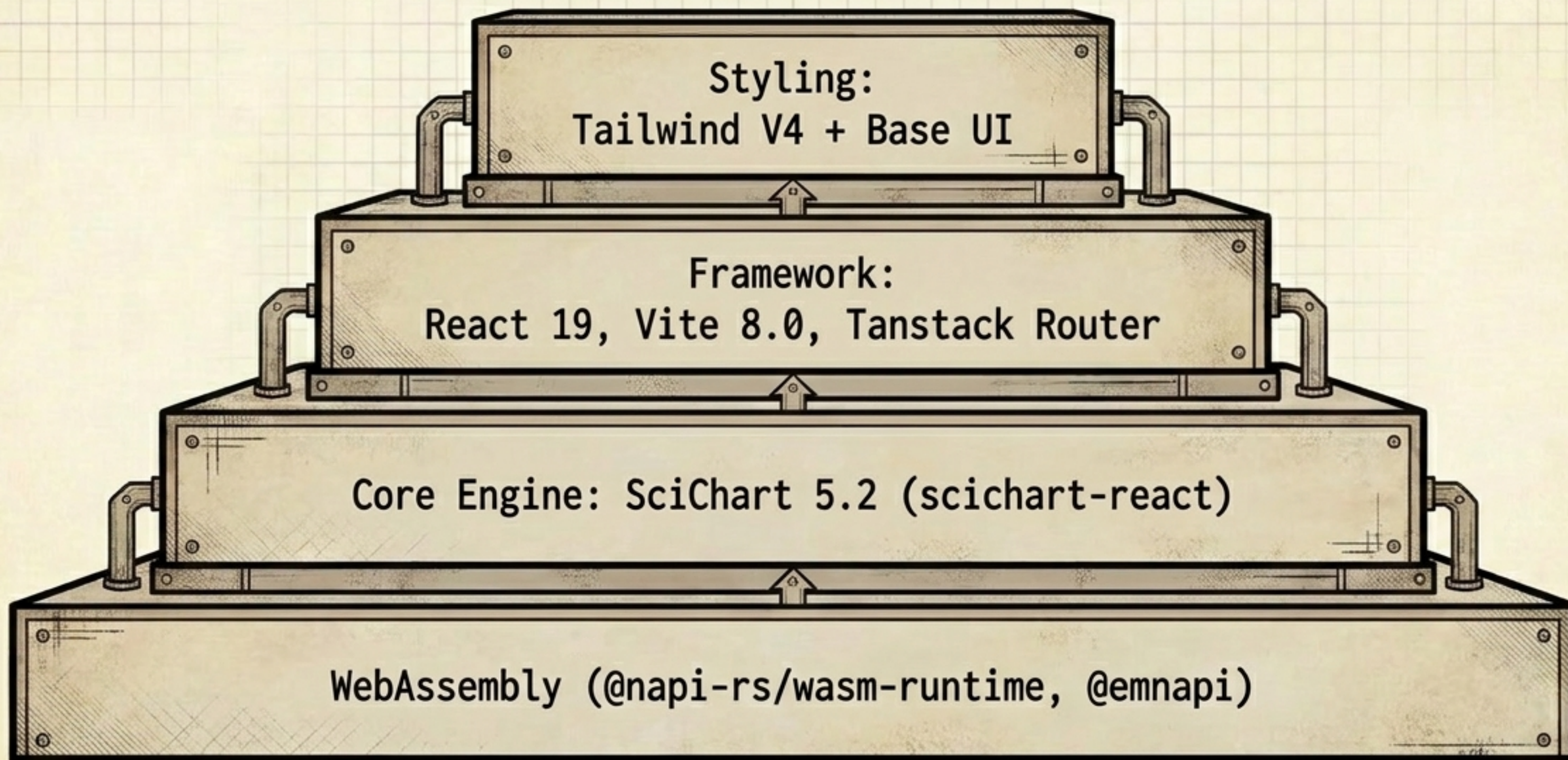
The **optimizer** uses **beam search** across **historical captures** to discover robust rule trees, demanding out-of-sample survival before deployment.

holdout_fraction: 0.25
max_holdout_decay: 0.5
(Ensures candidate forest survives out-of-sample)



THE HIGH-PERFORMANCE TELEMETRY STACK

The UI relies on a **WebAssembly-powered stack** to render hundreds of thousands of datapoints per second without collapsing under orderbook physics data.



THE CYBERNETIC LOOP: FROM METAL TO WEBGL

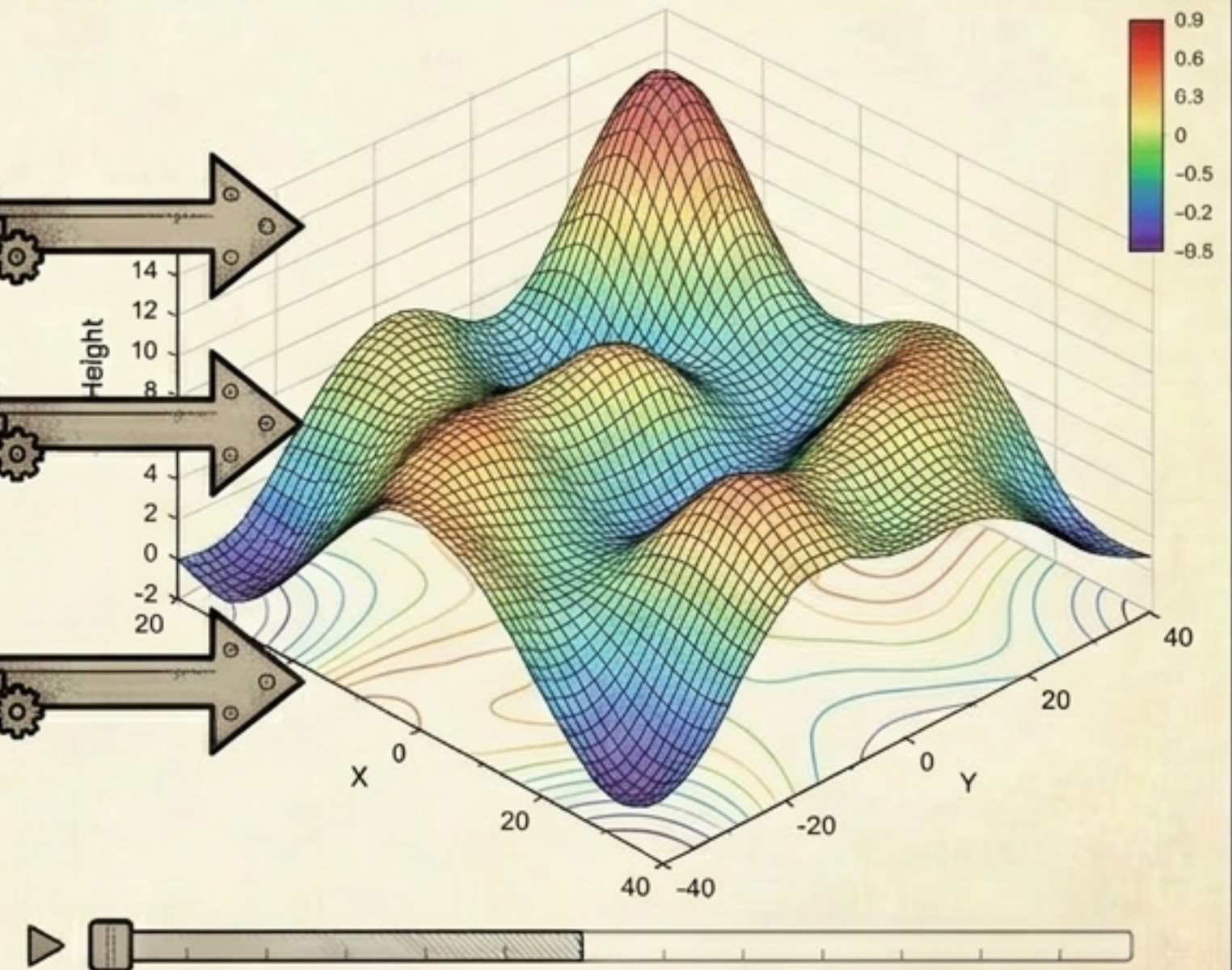
Symmetry between **computation** and **observation**. Continuous physics calculated in backend libraries are perfectly mirrored by 3D rendering engines in the frontend browser.

numeric/physics/manifold.metal

```
// Backend Physics Calculation
struct ManifoldParams {
    double k_amplitude = 1.5; // Surface Height
    double w_frequency = 2.0; // Wave Density
    double d_damping = 0.95;  // Decay Factor
    double t_time;           // Temporal Dimension
};

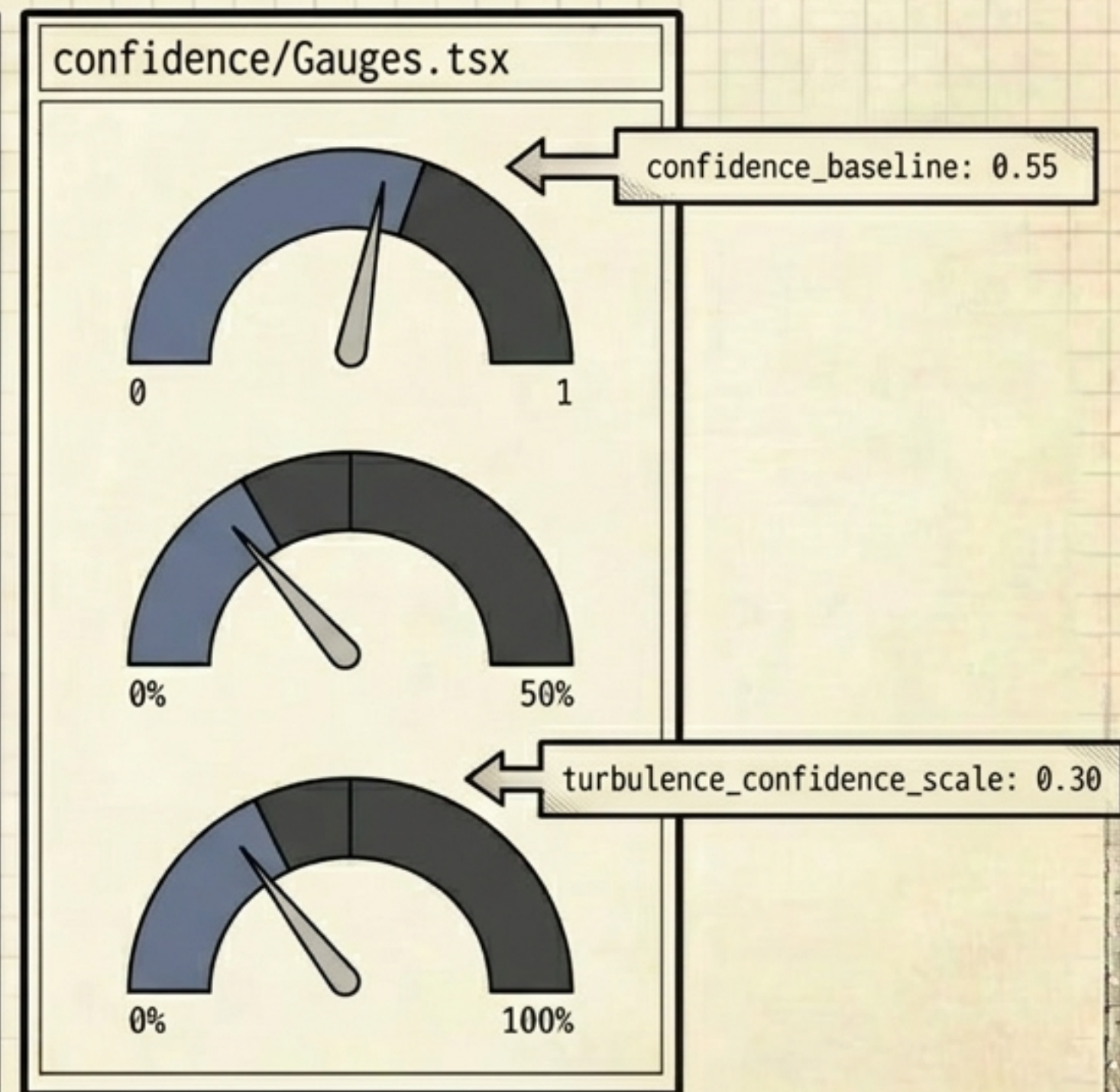
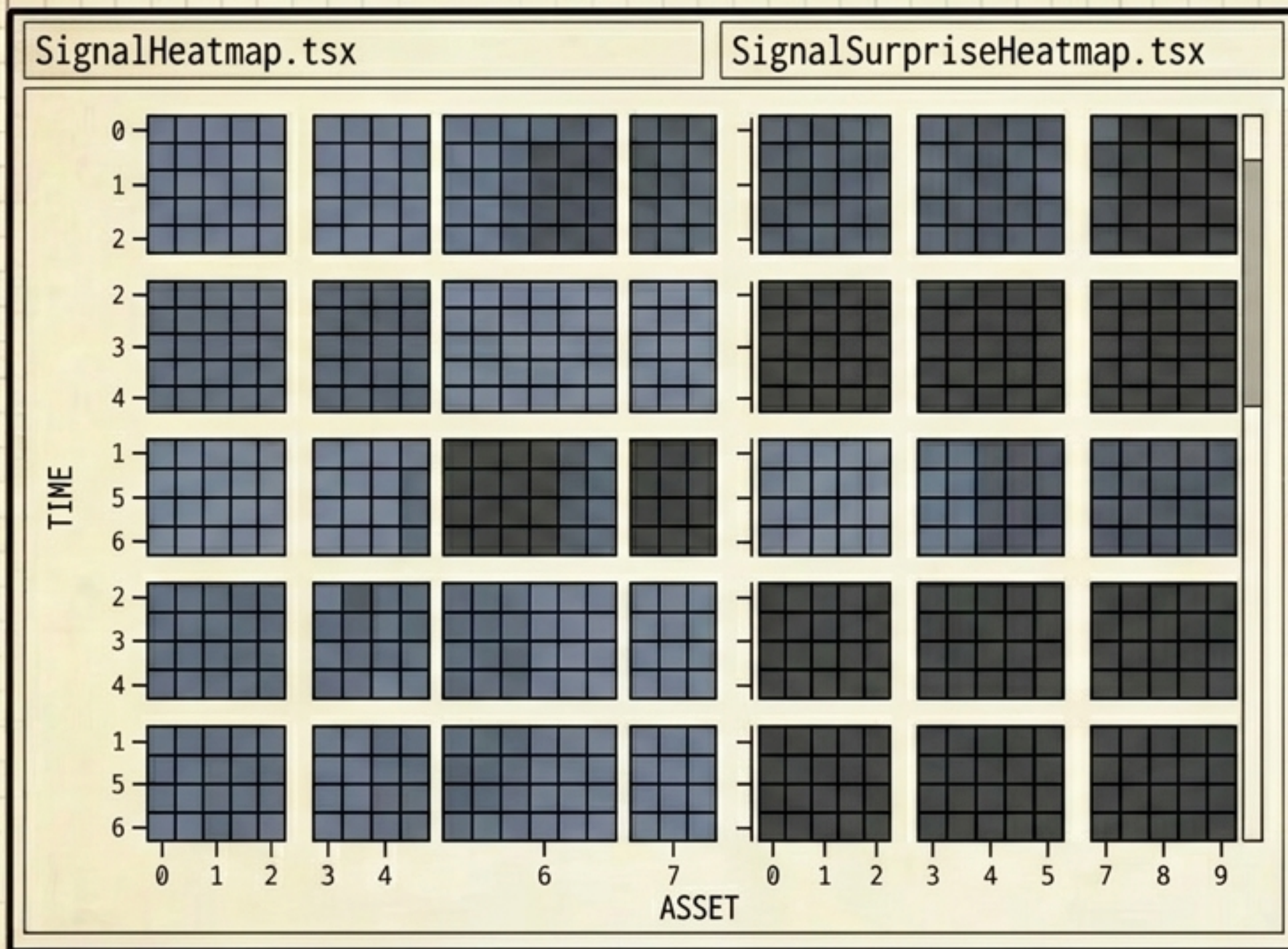
void computeManifold() {
    // Solve partial differential equations...
    // Calculate vertex positions and normals...
}
```

frontend/src/.../manifold/ManifoldSurfaceChart.tsx



DIAGNOSTIC AND DECISION DASHBOARDS

Exposing confidence gauges, signal heatmaps, and surprise thresholds to provide operators with immediate structural insight.



Symm: Operating at the Edge of Chaos

A unified architecture bridging deterministic logic, continuous math, and human oversight.

